

RESEARCH ARTICLE

WILEY

Influence of peer information and personality on social exclusion in a three-person dictator game

Manfred Königstein | Miriam Zschoche  | Tashauna A. Brown | Moritz Königstein

University of Erfurt, Erfurt, Germany

Correspondence

Miriam Zschoche, University of Erfurt,
Nordhaeuser Str. 63, 99089 Erfurt, Germany.
Email: miriam.zschoche@uni-erfurt.de

Funding information

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Abstract

We study the conflict between financial interests and prosocial behavior in a new variant of a three-person dictator game, which we refer to as “social exclusion game.” In this game, one person, the dictator, decides whether all three players (the dictator and two receivers) receive the fair outcome (equal payoffs) or an unfair outcome, which excludes one of the receivers and raises the dictator's payoff. We elicit the minimal payment necessary to induce the dictator to exclude the third person and investigate how this is influenced by peer information and the Big Five personality factor agreeableness. The data support our hypotheses.

1 | INTRODUCTION

Economic decisions quite often involve a conflict between financial interests and social norms or, more generally, prosocial behavior. This is obvious when firms decide to downsize and lay off employees or whether or not to establish sustainable business practices. It can also be seen in laboratory experiments on ultimatum games (Güth et al., 1982) and dictator games (Forsythe et al., 1994; Kahneman et al., 1986), which induce a conflict between financial self-interest and norms of fairness (Diaz et al., 2023; Engelmann & Strobel, 2010; Kerschbamer & Müller, 2020).¹ A number of theoretical models (social preference models) have been developed to give a formal representation of such issues (e.g., Bolton & Ockenfels, 2000; Charness & Rabin, 2002; Dufwenberg & Kirchsteiger, 2004; Falk & Fischbacher, 2006; Fehr & Schmidt, 1999).

An interesting question is whether and to what degree the conflict between financial self-interest and prosocial behavior is influenced by peer information and personality characteristics of the decision maker. We study this issue in a new variant of a three-person dictator game which we refer to as “social exclusion game.” In this game, one person, the dictator, decides whether all three players (the dictator and two receivers) receive the fair outcome (equal payoffs) or an unfair outcome, which excludes one of the receivers and raises the dictator's payoff.² Specifically, the dictator decides for a menu of 23 such binary choices in which the decision to exclude becomes more and more financially attractive for the dictator. This elicits each individual's minimal payment necessary to decide for exclusion of the

third person.³ We refer to it as variable *switch* for the switch from non-exclusion to exclusion. Since one of the 23 situations is randomly chosen and paid, the elicitation of *switch*, and thus the conflict between financial interest and prosocial behavior, is implemented in an incentive-compatible manner.

Excluding others from receiving any payoff is a severe action that may create cognitive dissonance within a decision maker (Carpenter et al., 2005) and allows for peer influence. To study the effect of peer information we apply two treatments: WI (with information) and NI (no information). Specifically, we used the distribution of choices observed in some NI sessions to determine the 20% percentile value of *switch*. This value informs that 20% of the dictators are willing to exclude even if they gain only 0.10 EUR (while the excluded person loses 2.00 EUR). In treatment WI, we communicate this to the participant. So, our peer information condition informs about the behavior of the 20% most egoistic participants. We expect that such peer information shifts the distribution of *switch* to the left inducing informed dictators to exclude at lower levels of payoff compensations (Hypothesis 1).

Furthermore, we collect the big five personality characteristics (Borkenau & Ostendorf, 1993; Costa & McCrae, 1992; McCrae & Costa, 1989). We expect that the personality factor agreeableness correlates negatively with the willingness to exclude the third person (Hypothesis 2). Namely, agreeableness is seen as relevant in social interactive situations, and persons high on this factor are more altruistic, trusting, nurturing, others-centered, etc. (McCrae & John, 1992). Both Hypotheses 1 and 2 are supported by the data.

In addition to the social exclusion game, our experiment comprised a hypothetical business case: The participants are asked under what financial circumstances they would decide for layoffs if they were manager of the described firm. We intend to study the influence of peer information and personality also in the hypothetical business case. However, since these decisions are non-incentivized, the participants may underreport their willingness to lay off. Actually, a large fraction of participants refused laying off under any of the given circumstances. Therefore, these data are unreliable.⁴ We only analyze them for explorative reasons, and we report the exact procedures of the hypothetical business case, the data received, and the analyses only in Appendix A.

We proceed as follows. In Section 2, we develop our hypotheses. In Section 3, we explain our empirical analysis in detail before we present the results in Section 4. Section 5 presents the discussion of findings, including their limitations and avenues for future research.

2 | HYPOTHESES

Studying the influence of peers on individual decision making is inherently relevant for understanding the general dynamics of human social interactions. Research shows that peer behavior has a strong influence when making moral decisions, for the good and for the bad: people are more likely to be cooperative and helpful when others are (e.g., Dimant, 2019; Nook et al., 2016) and punish and harm others if peers also do it (e.g., Fabbri & Carbonara, 2017; Yu et al., 2021). These contagion effects in (moral) decision making can be ascribed to psychological, sociological, and evolutionary mechanisms (Baddeley, 2010). As a social learning process, imitation enables rapid transmission of ideas and strengthens group coherence. Especially, when uncertainty is high, individuals are aware of their own ignorance and tend to mimic the decisions of others as they are expected to be better informed (Baddeley, 2010).

Information about peer behavior is typically available in many natural settings, where individuals do not act in social isolation. However, most laboratory studies using dictator-game-like settings focus on tightly controlled, but asocial decision environments, where individuals face neutral and abstract decision situations, under anonymity, and in complete isolation from other decision makers (e.g., Eckel & Grossman, 1996; Krupka & Weber, 2009, 2013). One exception that explicitly investigates the interdependence of individual decision making and peer behavior is the study of Gächter et al. (2017) who present a set of dictator game experiments that measure the influence of peers on both actual sharing and norms of sharing. Results show that the presence of peers has a systematic and strong influence on the perceptions of social appropriateness (i.e., social norms). However, there is no clear correlation between peer information and actual sharing behavior. This points to a nuanced relationship between individual decisions and the decisions of others.

When considering exclusion of others, that is, a proposed split where one person gets nothing, we expect that decision makers tend to choose this option when they know that others did so too. Unequal

distributions cause cognitive dissonance and violate social norms (Diaz et al., 2023). Knowing that other decision makers took that decision may serve as a legitimization and rationalization for individual behavior that excludes others for one's own benefit. Hence, our first hypothesis reads as:

Hypothesis 1. A decision maker is more willing to exclude a third player if she/he has the information that other decision makers also decided for exclusion.

In a second step, we also want to disentangle if there is an idiosyncratic propensity of human beings to exclude others per se. Previous research underpins the importance of a decision maker's personality characteristics in decision making in the lab (e.g., Sabater-Grande et al., 2022) as well as in real-life business situations (e.g., Gupta et al., 2019).

The five-factor model of personality ("Big Five") is a very influential concept from psychology (Costa & McCrae, 1992; Digman, 1990) that is intensively used to study decision-making mechanisms in diverse research fields such as economics (e.g., Ben-Ner et al., 2004), finance (e.g., Oehler et al., 2018), and management (e.g., Herrmann & Nadkarni, 2014). Several studies (e.g., Zhao et al., 2017; Zhao & Smillie, 2015) reveal that personality characteristics measured by the Big Five characteristics can predict a person's tendency toward a fair split. The inclination to exclude others, however, is not explicitly considered.

The Big Five are the factors *neuroticism*, *extraversion*, *openness to experience*, *conscientiousness*, and *agreeableness*. Depending on the specific decision situation, different factors can be the dominant driver for directing human behavior. Regarding the decision to exclude others, we believe that a decision maker's *agreeableness* is of particular relevance because of the following reasons.

Agreeableness is characterized by a positivity toward others, such that those people scoring higher in *agreeableness* are more altruistic, trusting, nurturing, others-centered, etc. (McCrae & John, 1992). It is strongly positively related to values of benevolence and strongly negatively related to values of power, suggesting that high *agreeableness* individuals seek environments and opportunities where they can help and serve others (Roccas et al., 2002). Given the socially oriented nature of this personality dimension, it is not surprising that students higher on *agreeableness* perform better in environments that foster social learning opportunities (Pawlowska et al., 2014). And while *agreeableness* is only weakly positively associated with overall educational aspirations, it is strongly positively associated with social vocational interests and with confidence in one's social skills (Rottinghaus et al., 2005). Individuals higher on *agreeableness* are less likely to be entrepreneurs, presumably since being an entrepreneur requires that people prioritize their own interests and success and that they not be overly naïve or trusting; as such, it makes sense that more agreeable individuals would not seek out such an environment (Zhao & Seibert, 2006). Similarly, *agreeableness* is negatively associated with risk taking in other domains, for example, risky recreational activities, health and safety behaviors, career and financial decisions, and social

situations (Nicholson et al., 2005). Under conditions of emotional exhaustion, individuals with lower *agreeableness* become more likely to engage in deviant behaviors (i.e., behaviors that could harm other people or organizations). On the other hand, individuals higher on *agreeableness* are less affected by emotional exhaustion (at least in terms of how it affects their engagement in deviant behavior), presumably because they are deterred by the possibility of doing something that would negatively affect others (Golparvar et al., 2014).

Several studies have now shown that *agreeableness* is a consistent positive predictor of prosocial game behaviors as well as dictator offers (e.g., Baumert et al., 2014; Becker et al., 2012; Ben-Ner & Kramer, 2011). Accordingly, we expect that lower values of *agreeableness* will enhance a decision maker's inclination to exclude others. Our second hypothesis reads as:

Hypothesis 2. A decision maker is more willing to exclude a third player if she/he has a lower value of *agreeableness*.

3 | EXPERIMENTAL DESIGN AND PROCEDURES

We conducted an experimental study in December 2019 in the computerized experimental laboratory of a university based in Germany using the computer software Z-tree v4.1.9 with 205 participants. Session cohorts varied between 11 and 25 participants. All participants were university students with different educational backgrounds, out of which 155 were females and 45 males and five refused to give information on gender. Mean age was 22.1 (std. dev. 4.3). All instructions were given on screen. (screenshots are available upon request.) The participants received a fixed payment for participation plus a payment that varied depending on decisions in the experiment. A session took about 45 min, and average earnings were about 12 EUR, which was more than the hourly wage for a student job (at the time of the experiment). In addition to the incentivized social exclusion game, participants were asked to make decision in a hypothetical business case (detailed description in the Appendix). Further, we collected personality data using the German version of the NEO-FFI according to Borkenau and Ostendorf (1993) as well as some socio-demographics (age, gender, field of study). In the following, we will describe the design of the social exclusion game in detail.

Consider a three-person game, in which player X, the “dictator,” may decide for an unequal payoff vector (250, 200, 0) that gives 250 for himself/herself, 200 for player Y, and 0 for player Z or an equal payoff vector (200, 200, 200). If player X decides for the unequal payoff vector, he/she is said to “exclude” player Z because this player gets zero. In a stochastic variant of this game, player X sees a table with 23 rows, each of which shows an equal payoff vector (200, 200, 200) on the left and an unequal payoff vector (x, 200, 0) on the right (see Table 1).⁵ The payoff x increases across rows starting with 190 in row 1 and ending with 550 in row 23. So, social exclusion

TABLE 1 Payoffs in the social exclusion game.

Row	Combination 1	Combination 2
1	Player X: 200; Player Y: 200, Player Z: 200	Player X: 190; Player Y: 200, Player Z: 0
2	Player X: 200; Player Y: 200, Player Z: 200	Player X: 200; Player Y: 200, Player Z: 0
...	...	...
13	Player X: 200; Player Y: 200, Player Z: 200	Player X: 310; Player Y: 200, Player Z: 0
14	Player X: 200; Player Y: 200, Player Z: 200	Player X: 330; Player Y: 200, Player Z: 0
...	...	...
19	Player X: 200; Player Y: 200, Player Z: 200	Player X: 430; Player Y: 200, Player Z: 0
20	Player X: 200; Player Y: 200, Player Z: 200	Player X: 460; Player Y: 200, Player Z: 0
...	...	...
23	Player X: 200; Player Y: 200, Player Z: 200	Player X: 550; Player Y: 200, Player Z: 0

is financially increasingly attractive across rows. An egoistic, income-maximizing, player should decide for the equal payoff vector in row 1 and for the unequal vector in rows 2 (3) to 23. However, if a player is socially motivated, he/she should switch to the unequal vector on a higher row number. Players are instructed to decide up to which row he/she chooses the equal payoff vector, implying consequently that from the next row on he/she switches to the unequal vector, that is, to social exclusion. To take a choice, a participant simply entered an integer between (and including) 0 and 23 with 0 (23), meaning that always, the unequal (equal) vector is chosen. We refer to this choice variable as the variable *switch*.

After choosing *switch*, a random draw determines the row number, which is relevant for payment. It is determined whether on the drawn row player 1 decided for the equal or unequal payoff vector and payments are made to the three players accordingly. This concludes the game. After playing the social exclusion game in the role of player 1, two more social exclusion games were announced and run, in which each participant had the role of players 2 and 3. Of course, there is no choice to be taken in these two games; however, by randomly matching players in groups of three, one role each, we truthfully paid the participants according to player 1's decision.

Furthermore, we introduced peer information. The game is similar to two-person dictator games studied by Blanco et al. (2011) as well as Dannenberg et al. (2007). In both of these studies, the games were used to determine individual fairness parameters to be used for predictions across different games. In our case, we simply want to study under what circumstances players take the antisocial decision to exclude one player out of a group of three and how this is affected by peer behavior. Namely, in one treatment condition, we give the experimental participants the following (true) information:

In a recent study on this experiment 20% of the participants decided in rows 1 to 3 for combination 1 [the equal payoff vector–authors] and in row 4 the first time for combination 2 [the unequal payoff vector–authors].

We refer to this as treatment WI (with information) and compare it to treatment NI (no information), in which we did not include this sentence. To provide the peer information truthfully, we retrieved it from two NI-sessions, which we ran first. According to Hypothesis 1, the average value of *switch* should be lower in WI than in NI.

Note that we do not investigate whether an exclusion decision may be distorted by biased peer information. Rather, we investigate whether it is influenced by truthful peer information regarding the 20% percentile of individuals most willing to exclude.

4 | RESULTS

Figure 1 displays a cumulative distribution of the experimental choice variable *switch*, which identifies the row up to which a participant chooses option 1, switching to option 2 from the next row on. The figure shows, for example, that in treatment WI, about 40% of the participants switch on row 3. That is, they exclude the third player when the unequal payoff vector is (220, 200, 0), offering a payoff of 220 or higher for themselves. In contrast, in treatment NI, about 40% of the participants switch on row 12, thereby excluding the third player when the unequal payoff vector is (310, 200, 0), offering 310 or more for themselves. In both treatments, a substantial number of players decided not to exclude at all by choosing *switch* = 23. This

implements equal payoffs of 200 for each player under all circumstances implying that players forgo a payoff of 550 instead of 200 for themselves. Such strongly prosocial behavior was chosen by 41.0% (41/100) of the players in NI and by 22.9% (24/105) in WI.

Table 2 shows the means and standard deviations for the different treatments. The difference in *switch* between treatments is significant according to a Mann–Whitney *U* test ($p = 0.005$, two-tailed, $N = 205$, Cohen's $d: 9.145$ [large effect size]). According to the medians, an average experimental participant excludes the third player in NI if the payoff vector is (410, 200, 0), that is, for a personal benefit of 2.10 EUR. In the WI condition, the average player does so if the payoff vector is (260, 200, 0), that is, if the personal benefit is 0.60 EUR.

To investigate whether the willingness to exclude correlates with personality, we regress *switch* on the five personality factors in addition to a dummy for the information treatment. The five factors are intercorrelated as shown in Table 3. All 10 reported intercorrelations fit qualitatively with those reported in Borkenau and Ostendorf (1993) (Table 3) for a gender-mixed sample. Most of them are statistically significant. Therefore, we have to consider multicollinearity in regression analyses. Since the dependent variable *switch* is bounded above and below with a substantial number of observations (65) at the upper bound *switch* = 23 (and 18 at the lower bound 0), we ran Tobit regression analyses and report the results in Table 4.

TABLE 2 Descriptive statistics for variable *switch*.

	Median	Mean	Std. dev.	Obs.
NI: Without information	17	13.85	9.37	100
WI: With information	7	9.97	8.92	105

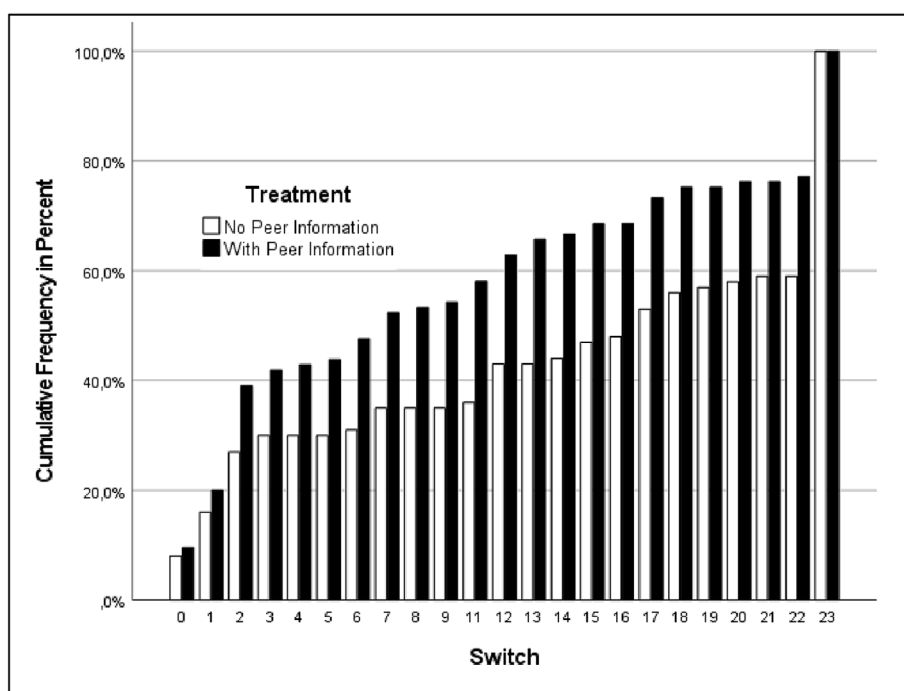


FIGURE 1 Cumulative frequency distribution for variable *switch*.

	Neuroticism	Extraversion	Openness	Agreeableness
Neuroticism	1.000			
Extraversion	−0.335**	1.000		
Openness to experience	0.141*	0.138*	1.000	
Agreeableness	−0.102	0.401**	0.152*	1.000
Conscientiousness	−0.241**	0.172*	−0.148*	0.054

Note: Asterisks indicate two-tailed significance.

* $p < 0.05$, and ** $p < 0.01$.

TABLE 3 Measured intercorrelations of personality factors.

Variable	Model 1			Model 2		
	Coeff.	Std. err.	P-value	Coeff.	Std. err.	P-value
Constant	22.88	10.97	0.038	7.50	5.21	0.152
Peer information	−6.15	2.09	0.004	−6.02	2.09	0.004
Agreeableness	4.76	2.21	0.033	3.90	2.03	0.056
Extraversion	−2.05	2.23	0.361			
Openness to experience	−0.97	2.08	0.641			
Neuroticism	−1.95	1.85	0.293			
Conscientiousness	−1.71	1.83	0.350			
Model	N = 205 Pseudo $R^2 = 0.013$ $p = 0.021$			N = 205 Pseudo $R^2 = 0.010$ $p = 0.002$		

Note: P-values refer to two-tailed coefficient tests.

TABLE 4 Tobit regression analyses. Dependent variable: *switch*.

Model 1 in Table 4 shows regression results when all five personality factors are entered in addition to a treatment dummy *peer information* (0 = NI, 1 = WI). Model 2 shows the regression result when only the factor *agreeableness* is entered in addition to *peer information*. As predicted, *peer information* has a negative effect and *agreeableness* has a positive effect. These two variables show significant influences in both regression models supporting Hypothesis 1 and Hypothesis 2. That is, experimental participants are more willing to exclude a player if they are informed that other participants have done so and if they score lower on the factor *agreeableness*. Sequential backward elimination of insignificant regressors of the full model (1) leads to the restricted model (2). The overall fit of the Tobit model, however, indicated by the pseudo R^2 is weak. As an additional robust measure, we determine the Spearman rank correlation coefficient between *switch* and *agreeableness*. It is $\rho = 0.146$ and it is significant ($p = 0.037$, two-tailed, $N = 205$). Summarizing, we find that the influences of peer information and *agreeableness* are as predicted by Hypothesis 1 and Hypothesis 2. While the influence of peer information is as rather strong, the influence of *agreeableness* is weak.

5 | SUMMARY AND DISCUSSION

The goal of this study was to shed light on the question whether the decision to exclude others is determined by peer behavior and a decision maker's personality. We study this questions in a new variant of a three-person dictator game which we refer to as “social exclusion

game.” In this game, one person, the dictator, decides whether all three players (the dictator and two receivers) receive equal payoffs or an unequal outcome, which excludes one of the receivers and raises the dictator's payoff. A laboratory study with 205 student participants revealed influences as predicted by our hypotheses: The decision to exclude is taken more easily if it follows peer examples and if the decision maker scores low on personality factor *agreeableness*.

Being informed that 20% of players exclude a third player for a negligible personal benefit of 0.10 EUR or less induces a substantial change in average behavior. The median value of the decision variable *switch* shifts from 17 in NI to 7 in WI; that is, the personal benefit necessary to induce social exclusion shifts from 2.10 EUR to 0.60 EUR. The influence of peer information is significant according to (robust) nonparametric tests and according to Tobit regression analyses, which supports Hypothesis 1. Further, the analyses also showed a significant effect of the personality factor *agreeableness* on the inclination to exclude others, which supports Hypothesis 2. Participants scoring high on this factor are considered more prosocial in interactive environments; our findings reveal a reduced willingness to exclude others of those people.

In a supplementary analysis (detailed description can be found in the Appendix), we framed a hypothetical business decision scenario. Laying off employees can be interpreted as exclusion of others; it causes moral costs, which decision makers try to reduce (Keum & Meier, 2023). We asked participants to take the role of a manager and to make a layoff decision. In this design, decision making is not financially incentivized. The results, however, did not match those of the

incentivized exclusion game. We found support neither for our first hypothesis nor for the second hypothesis. We offer explanations for those findings in the Appendix.

The results of this study complement previous research that investigates under what conditions human beings act more prosocial when sharing a resource (e.g., Eckel & Grossman, 1996; Jang & Lynham, 2015). In this paper, we expect that the decision to exclude others completely creates severe cognitive dissonance for decision makers, which can be reduced when decision makers are informed that peers (other decision makers) also excluded others. Hence, our paper complements research that highlights the importance of peers in social interactions involving sharing decisions (Gächter et al., 2017). Further, our results add to existing research that identifies *agreeableness* as a factor predicting prosocial behavior in diverse laboratory settings (e.g., Ben-Ner & Kramer, 2011; Zhao et al., 2017).

The findings of this study may have some interesting implications for decision makers in firms. From time to time, firms must take decisions that exclude others, for example, when laying off employees. The study reveals that decision situations, in which social consideration have to be weighed against financial considerations, might be easier to handle for decision makers when looking at peer behavior. Indeed, we can observe in real life that firms downsize their workforce when other firms do so as well (Ahmadjian & Robinson, 2001). Our study delivers a potential explanation for this observation.

Another practical implication can be deduced from the result that the personality factor *agreeableness* eases a decision maker's tendency to exclude others. We see that, besides increasing professionalization in the management of organizations and attempts for transparent and comprehensible decisions, personal characteristics and attitudes can influence strategic decision making.

When interpreting the results of this study, one should keep its limitations in mind. We build on a student sample, which is common in behavioral economics research, but which might limit the generalizability of the findings. Still, we are confident that our results are universal in their nature. Most human beings strive to reduce cognitive dissonance when making decisions. Hence, we expect that personality characteristics and the tendency to align to peer behavior are influencing factors that are also present in decisions to exclude others made by, for example, experienced managers. Still, future research might use empirical setups that are closer to the real business life. Experiments with participants that have managerial positions in firms could be fruitful and promising extensions.

CONFLICT OF INTEREST STATEMENT

The authors have no relevant financial or nonfinancial interests to disclose.

DATA AVAILABILITY STATEMENT

The dataset is available from the corresponding author upon request.

ORCID

Miriam Zschoche  <https://orcid.org/0000-0001-8558-652X>

ENDNOTES

- ¹ Few studies investigate social preferences in real-life settings (e.g., Jang & Lynham, 2015).
- ² One real-world example of this situation might be the managerial decision to lay off employees in order to increase firm profit (Dwyer & Arbelo, 2012).
- ³ A similar technique to elicit dictator preferences was applied in Blanco et al. (2011) in a two-player dictator game.
- ⁴ We describe a business situation (see Appendix) that is a substantial threat for the survival of the firm and describe substantial possible improvements feasible by layoffs. We think that in real life, such situations would quite naturally lead to layoffs.
- ⁵ In the experiment, players were labeled "participant X," "participant Y," and "participant Z."

REFERENCES

- Ahmadjian, C. L., & Robinson, P. (2001). Safety in numbers: Downsizing and the deinstitutionalization of permanent employment in Japan. *Administrative Science Quarterly*, 46(4), 622–654. <https://doi.org/10.2307/3094826>
- Baddeley, M. (2010). Herding, social influence and economic decision-making: Socio-psychological and neuroscientific analyses. *Philosophical Transactions of the Royal Society, B: Biological Sciences*, 365(1538), 281–290. <https://doi.org/10.1098/rstb.2009.0169>
- Baumert, A., Schlösser, T., & Schmitt, M. (2014). Economic games: A performance-based assessment of fairness and altruism. *European Journal of Psychological Assessment*, 30, 178–192. <https://doi.org/10.1027/1015-5759/a000183>
- Becker, A., Deckers, T., Dohmen, T., Falk, A., & Kosse, F. (2012). The relationship between economic preferences and psychological personality measures. *Annual Review of Economics*, 4, 453–478. <https://doi.org/10.1146/annurev-economics-080511-110922>
- Ben-Ner, A., & Kramer, A. (2011). Personality and altruism in the dictator game: Relationship to giving to kin, collaborators, competitors, and neutrals. *Personality and Individual Differences*, 51(3), 216–221. <https://doi.org/10.1016/j.paid.2010.04.024>
- Ben-Ner, A., Putterman, L., Kong, F., & Magan, D. (2004). Reciprocity in a two-part dictator game. *Journal of Economic Behavior & Organization*, 53(3), 333–352. <https://doi.org/10.1016/j.jebo.2002.12.001>
- Blanco, M., Engelmann, D., & Normann, H. T. (2011). A within-subject analysis of other-regarding preferences. *Games and Economic Behavior*, 72(2), 321–338. <https://doi.org/10.1016/j.geb.2010.09.008>
- Bolton, G., & Ockenfels, A. (2000). ERC: A theory of equity, reciprocity, and competition. *American Economic Review*, 90, 166–193. <https://doi.org/10.1257/aer.90.1.166>
- Borkenau, P., & Ostendorf, F. (1993). NEO-Fünf-Faktoren-Inventar (NEO-FFI) nach Costa und McCrae: Manual.
- Carpenter, J., Verhoogen, E., & Burks, S. (2005). The effect of stakes in distribution experiments. *Economics Letters*, 86(3), 393–398. <https://doi.org/10.1016/j.econlet.2004.08.007>
- Charness, G., & Rabin, M. (2002). Understanding social preferences with simple tests. *Quarterly Journal of Economics*, 117(3), 817–869. <https://doi.org/10.1162/003355302760193904>
- Costa, P. T., & McCrae, R. R. (1992). Normal personality assessment in clinical practice: The NEO personality inventory. *Psychological Assessment*, 4(1), 5–13. <https://doi.org/10.1037/1040-3590.4.1.5>
- Dannenberg, A., Riechmann, T., Sturm, B., & Vogt, C. (2007). Inequity aversion and individual behavior in public good games: An experimental investigation. ZEW-Centre for European Economic Research Discussion Paper, 07-034. <https://doi.org/10.2139/ssrn.991555>

- Diaz, L., Houser, D., Ifcher, J., & Zarghamee, H. (2023). Estimating social preferences using stated satisfaction: Novel support for inequity aversion. *European Economic Review*, 155, 104436. <https://doi.org/10.1016/j.eurocorev.2023.104436>
- Digman, J. M. (1990). Personality structure: Emergence of the five-factor model. *Annual Review of Psychology*, 41(1), 417–440. <https://doi.org/10.1146/annurev.ps.41.020190.002221>
- Dimant, E. (2019). Contagion of pro-and anti-social behavior among peers and the role of social proximity. *Journal of Economic Psychology*, 73, 66–88. <https://doi.org/10.1016/j.joep.2019.04.009>
- Dufwenberg, M., & Kirchsteiger, G. (2004). A theory of sequential reciprocity. *Games and Economic Behavior*, 47, 268–298. <https://doi.org/10.1016/j.geb.2003.06.003>
- Dwyer, D. J., & Arbelo, M. (2012). The role of social cognition in downsizing decisions. *Journal of Managerial Psychology*, 27(4), 383–405. <https://doi.org/10.1108/02683941211220199>
- Eckel, C. C., & Grossman, P. J. (1996). Altruism in anonymous dictator games. *Games and Economic Behavior*, 16(2), 181–191. <https://doi.org/10.1006/game.1996.0081>
- Engelmann, D., & Strobel, M. (2010). Inequality aversion and reciprocity in moonlighting games. *Games*, 1(4), 459–477. <https://doi.org/10.3390/g1040459>
- Fabrizi, M., & Carbonara, E. (2017). Social influence on third-party punishment: An experiment. *Journal of Economic Psychology*, 62, 204–230. <https://doi.org/10.1016/j.joep.2017.07.003>
- Falk, A., & Fischbacher, U. (2006). A theory of reciprocity. *Games and Economic Behavior*, 54(2), 293–315. <https://doi.org/10.1016/j.geb.2005.03.001>
- Fehr, E., & Schmidt, K. M. (1999). A theory of fairness, competition, and cooperation. *Quarterly Journal of Economics*, 114(3), 817–868. <https://doi.org/10.1162/003355399556151>
- Forsythe, R., Horowitz, J. L., Savin, N. E., & Sefton, M. (1994). Fairness in simple bargaining experiments. *Games and Economic Behavior*, 6, 347–369. <https://doi.org/10.1006/game.1994.1021>
- Gächter, S., Gerhards, L., & Nosenzo, D. (2017). The importance of peers for compliance with norms of fair sharing. *European Economic Review*, 97, 72–86. <https://doi.org/10.1016/j.eurocorev.2017.06.001>
- Giachetti, C., & Torrisi, S. (2018). Following or running away from the market leader? The influences of environmental uncertainty and market leadership. *European Management Review*, 15(3), 445–463. <https://doi.org/10.1111/emre.12130>
- Golparvar, M., Dehghan, S., & Mehdad, A. (2014). Relationship between emotional exhaustion and deviant behaviors: Moderating role of big five personality traits. *International Journal of Scientific Research in Knowledge*, 2(6), 285–296. <https://doi.org/10.12983/ijrsk-2014-p0285-0296>
- Gupta, A., Nadkarni, S., & Mariam, M. (2019). Dispositional sources of managerial discretion: CEO ideology, CEO personality, and firm strategies. *Administrative Science Quarterly*, 64(4), 855–893. <https://doi.org/10.1177/0001839218793128>
- Güth, W., Schmittberger, R., & Schwarze, B. (1982). An experimental analysis of ultimatum bargaining. *Journal of Economic Behavior and Organization*, 3, 367–388. [https://doi.org/10.1016/0167-2681\(82\)90011-7](https://doi.org/10.1016/0167-2681(82)90011-7)
- Haunschild, P. R., & Miner, A. S. (1997). Modes of interorganizational imitation: The effects of outcome salience and uncertainty. *Administrative Science Quarterly*, 42(3), 472–501.
- Haveman, H. A. (1993). Follow the leader: Mimetic isomorphism and entry into new markets. *Administrative Science Quarterly*, 38(4), 593–627. <https://doi.org/10.2307/2393338>
- Herrmann, P., & Nadkarni, S. (2014). Managing strategic change: The quality of CEO personality. *Strategic Management Journal*, 35(9), 1318–1342. <https://doi.org/10.1002/smj.2156>
- Jang, C., & Lynham, J. (2015). Where do social preferences come from? *Economics Letters*, 137, 25–28. <https://doi.org/10.1016/j.econlet.2015.09.004>
- Kahneman, D., Knetsch, J., & Thaler, R. (1986). Fairness and the assumptions of economics. *Journal of Business*, 59, 285–300. <https://doi.org/10.1086/296367>
- Kerschbamer, R., & Müller, D. (2020). Social preferences and political attitudes: An online experiment on a large heterogeneous sample. *Journal of Public Economics*, 182, 104076. <https://doi.org/10.1016/j.jpubeco.2019.104076>
- Keum, D. D., & Meier, S. (2023). License to layoff? Unemployment insurance and the moral cost of layoffs. *Organization Science*, forthcoming. <https://doi.org/10.1287/orsc.2022.16734>
- Krupka, E., & Weber, R. A. (2009). The focusing and informational effects of norms on pro-social behavior. *Journal of Economic Psychology*, 30(3), 307–320. <https://doi.org/10.1016/j.joep.2008.11.005>
- Krupka, E., & Weber, R. A. (2013). Identifying social norms using coordination games: Why does dictator game sharing vary? *Journal of the European Economic Association*, 11(3), 495–524. <https://doi.org/10.1111/jeea.12006>
- Lieberman, M. B., & Asaba, S. (2006). Why do firms imitate each other? *Academy of Management Review*, 31(2), 366–385. <https://doi.org/10.5465/amr.2006.20208686>
- McCrae, R. R., & Costa, P. T. (1989). Reinterpreting the Myers-Briggs type indicator from the perspective of the five-factor model of personality. *Journal of Personality*, 57(1), 17–40. <https://doi.org/10.1111/j.1467-6494.1989.tb00759.x>
- McCrae, R. R., & John, O. P. (1992). An introduction to the five-factor model and its applications. *Journal of Personality*, 60(2), 175–215. <https://doi.org/10.1111/j.1467-6494.1992.tb00970.x>
- Nicholson, N., Soane, E., Fenton-O'Creevy, M., & Willman, P. (2005). Personality and domain-specific risk taking. *Journal of Risk Research*, 8(2), 157–176. <https://doi.org/10.1080/1366987032000123856>
- Nook, E. C., Ong, D. C., Morelli, S. A., Mitchell, J. P., & Zaki, J. (2016). Pro-social conformity: Prosocial norms generalize across behavior and empathy. *Personality and Social Psychology Bulletin*, 42(8), 1045–1062. <https://doi.org/10.1177/0146167216649932>
- Oehler, A., Wendt, S., Wedlich, F., & Horn, M. (2018). Investors' personality influences investment decisions: Experimental evidence on extraversion and neuroticism. *Journal of Behavioral Finance*, 19(1), 30–48. <https://doi.org/10.1080/15427560.2017.1366495>
- Pawlowska, D. K., Westerman, J. W., Bergman, S. M., & Huelsman, T. J. (2014). Student personality, classroom environment, and student outcomes: A person–environment fit analysis. *Learning and Individual Differences*, 36, 180–193. <https://doi.org/10.1016/j.lindif.2014.10.005>
- Roccas, S., Sagiv, L., Schwartz, S. H., & Knafo, A. (2002). The big five personality factors and personal values. *Personality and Social Psychology Bulletin*, 28(6), 789–801. <https://doi.org/10.1177/0146167202289008>
- Rottinghaus, P. J., Day, S. X., & Borgen, F. H. (2005). The career futures inventory: A measure of career-related adaptability and optimism. *Journal of Career Assessment*, 13(1), 3–24. <https://doi.org/10.1177/1069072704270271>
- Sabater-Grande, G., García-Gallego, A., Georgantzis, N., & Herranz-Zarzoso, N. (2022). The effects of personality, risk and other-regarding attitudes on trust and reciprocity. *Journal of Behavioral and Experimental Economics*, 96, 101797. <https://doi.org/10.1016/j.socec.2021.101797>
- Yu, H., Siegel, J. Z., Clithero, J. A., & Crockett, M. J. (2021). How peer influence shapes value computation in moral decision-making. *Cognition*, 211, 104641. <https://doi.org/10.1016/j.cognition.2021.104641>
- Zhao, H., & Seibert, S. E. (2006). The big five personality dimensions and entrepreneurial status: A meta-analytical review. *Journal of Applied Psychology*, 91(2), 259–271. <https://doi.org/10.1037/0021-9010.91.2.259>
- Zhao, K., Ferguson, E., & Smillie, L. D. (2017). Individual differences in good manners rather than compassion predict fair allocations of wealth in

the dictator game. *Journal of Personality*, 85(2), 244–256. <https://doi.org/10.1111/jopy.12237>

Zhao, K., & Smillie, L. D. (2015). The role of interpersonal traits in social decision making: Exploring sources of behavioral heterogeneity in economic games. *Personality and Social Psychology Review*, 19(3), 277–302. <https://doi.org/10.1177/1088868314553709>

How to cite this article: Königstein, M., Zschoche, M., Brown, T. A., & Königstein, M. (2024). Influence of peer information and personality on social exclusion in a three-person dictator game. *Managerial and Decision Economics*, 45(5), 3434–3443. <https://doi.org/10.1002/mde.4209>

APPENDIX A

A.1 | Hypothetical business case part 1: Between-subject design

A.1.1 | Experimental design

In part 1, the participants received the following information on a hypothetical business scenario:

Imagine you lead a company that produces at two locations with 1000 employees each. You operate in a market in which there are five other competing companies. In the last 3 years, location A has generated profits that amount to a yearly rate of return on capital (ROC) of 5%, which is usual in the capital market. However, location B has generated losses that amount to an ROC of –2%. An ROC of 5% means that your company earned a profit of 500.000 Euro per 10 million of invested capital. An ROC of –2% means that your company made a loss of 200.000 Euro per 10 million of invested capital.

In the long term, the losses of location B threaten your whole company. In order to increase the ROC of location B, you may take the decision to lay off 500 employees at location B. You know that such a decision is seen negatively by the public. How large would the feasible ROC have to be minimally to make you take the decision of laying off 500 employees?

This completes the baseline description of the scenario except for the information manipulation regarding peer behavior and instruction how to enter an ROC (see below). Haunschild and Miner (1997) distinguish among three modes of imitation: frequency-based imitation (following the widespread adoption of practices or investments), trait-based imitation (following successful peers with traits such as size or status), and outcome-based imitation (following firms with salient positive results in the past, avoiding those with bad results). According to the frequency-based imitation-logic, a decision maker's willingness to downsize should be higher if other industry peers downsize workforce as well. Trait-based imitation suggests that a crucial strategic signal might be sent by the industry (or “market”) leader (Giachetti & Torrisi, 2018). The market leading firm is presumed to be not only the most successful (Lieberman & Asaba, 2006),

but also the most visible firm (Haunschild & Miner, 1997; Haveman, 1993). Following the strategic decision of this firm can legitimize critical strategic decisions, so a decision maker's willingness to downsize should be higher if the market leader does so. Since a market-leading firm usually has also been rather successful in the past, it is difficult to distinguish such an effect from outcome-based imitation. Consequently, in our experiment we implement only the frequency of peer behavior and information on the market-leading peer. We implemented four treatments—no information (NI), low frequency (LF), high frequency (HF), and market leader (ML)—by adding a single sentence after the baseline description given above in treatments LF, HF, and ML. The sentences are:

- Treatment LF: *You know that one of your competing companies has recently taken similar decisions.*
- Treatment HF: *You know that four of your competing companies have recently taken similar decisions.*
- Treatment ML: *You know that one of your competing companies, which is the market leader, has recently taken similar decisions.*

In the control treatment without peer information, no sentence was added. Information treatments NI, LF, HF, and ML were applied between subjects to about 25% of the participant cohort each. By comparing treatments NI versus LF, LF versus HF, and LF versus ML we have different tests of the same underlying hypothesis, which predicts a reduced ROC with peer information, since imitating others makes a negatively loaded decision easier and therefore requires a lower financial compensation. It hinges, of course, on successfully inducing a morally loaded decision conflict within the participants. It is not clear a priori whether this manipulation will work, given that the choice problem is hypothetical and non-incentivized.

Following the baseline information and the treatment implementation, the participants were instructed how to enter an ROC. Numbers were restricted to the range –2 to +6 and could have two decimals. In addition, participants could decline to lay off and not provide an ROC at all. To check on understanding of ROC, there was a control question asking to calculate the profit corresponding to an ROC of 10% when invested capital is 20 million euro. Only after answering the control question correctly, a participant could enter an answer on the main question.

A.1.2 | Results

In part 1, only 109 out of 205 participants (53.2%) entered an ROC (see Table A1). The rest, 96 out of 205 (46.8%), chose the “decline” option. So, almost half of the participants refused to take the hypothetical layoff decision at any ROC in the given range –2% to +6%. Relying on those 109 participants, who did provide an ROC, Table A1 shows means and standard deviations of ROC by treatments.

TABLE A1 Descriptive statistics for choices in the hypothetical business case.

Treatment	ROC provided			ROC not provided	
	N	Mean	Std. dev.	N	Total N
NI	28	1.45	2.79	22	50
LF	34	2.37	2.82	16	50
HF	25	3.97	2.40	25	50
ML	22	2.95	2.77	33	55
	109	2.62	2.82	96	205

Contrary our expectations, the mean ROC is smaller without peer information (treatment NI) than with peer information (treatments LF, HF, and ML). It is also smaller in LF than in HF and smaller in LF than in ML.

The difference in ROC between NI and the pooled treatments with peer information is significant according to a Mann–Whitney *U* test ($p = 0.008$, two-tailed, $N = 109$). This finding does not seem to be driven by different frequencies of declining an ROC, since these are almost equal: 44.0% (22/50) in treatment NI and 47.7% (74/155) in the pooled treatments with peer information.

Table A2 reports Tobit regression analyses with ROC as dependent variable with lower bound -2% and upper bound $+6\%$. It relies on 109 cases in which an ROC was provided. Model 1 uses dummies for the different information treatments (treatment NI being the reference category) and the five personality factors measured by the NEO-FFI as explanatory variables. Model 2 in Table A2 excludes all personality factors except *agreeableness*.

The displayed effects of peer information in models 1 and 2 are in line with the above findings of non-parametric testing and they are counter to what we expected. The effect of *agreeableness* is insignificant. Overall, the data on the hypothetical business case collected in part 1 do not support our hypotheses.

A.2 | Hypothetical business case part 2: Within-subject design

A.2.1 | Experimental design

In part 2, the participants were asked to consider the same hypothetical business case as in part 1 but now for the three information treatments other than the one in part 1. Thus, a participant of the NI group in part 1 was asked for conditions LF, HF, and ML in part 2. A participant of the LF group in part 1 was asked for conditions NI, HF, and ML in part 2, and so on. Accordingly, by combining the data of parts 1 and 2, we have responses by each participant to all information conditions and may apply within-subject tests on treatment influences. We collected the part 2 data only for explorative reasons. Importantly, when answering questions on part 2, the participants could not go back to change answers on part 1. Part 2 was implemented by first referring to the own previous choice made by the participant in part 1 and then asking whether and how the decision would change given different information. An ROC (or decline choice) had to be provided for the remaining three information conditions.

A.2.2 | Results

The part 2 data were collected for explorative reasons allowing for within-subject analyses. The part 1 data revealed two puzzling findings: (1) the large frequency of decline decisions and (2) that peer information on average even increased the ROC. We offer two post hoc explanations for these findings: First, a number of participants felt the moral cost of laying off 500 employees too large to trade it off against an ROC increase in the offered range (-2% to $+6\%$). Hence, they rather chose the decline option. Second, among participants willing to lay off at some rate in the given range, there are some who consider the economic problem more severe and the moral cost of laying off even higher if they know that other firms have taken similar decisions (peer information).

To elaborate these ideas further, we define different types of participants and measure their relative frequency: Type A is a participant who obeys all of the order relations $NI \geq LF, LF \geq HF, LF \geq ML$, and one of these inequalities holding strongly. By $NI \geq LF$, we mean that the submitted ROC in treatment NI is larger than the one in LF. Other inequalities have analogous meaning. Type A is a player who trades off layoffs against reasonable ROC improvements.

Type B is a participant who obeys all of the order relations $NI \leq LF, LF \leq HF, LF \leq ML$, and one of these inequalities holding strongly. This player trades off layoffs against reasonable ROC improvement but feels that peer information increases the moral cost of laying off. Type C is a player who chooses the maximal ROC of 6% or the decline option in all treatments. It is a player who strongly rejects trading off layoffs against ROC improvements. Type D is a player who chooses the same ROC below 6% in all treatments. Both types C and D are uninformative regarding the influence of peer information. Type E is a residual class of players with inconsistent reaction to peer information, for example, $NI < LF, LF > HF$.

Using these categories, Table A3 displays relative frequencies of the different player types. The third column indicates whether or not the respective player type supports Hypothesis 1, stating that the information that peers also exclude others (lay off employees) leads to a higher inclination of the decision maker to exclude, too. While the majority of participant are classified as neutral, there is a majority of pro-types compared to contra-types among those who are non-neutral. Altogether, the post hoc analysis delivers some interesting additional insight, but does not support our hypotheses.

TABLE A2 Tobit regression analyses.
Dependent variable: ROC.

Variable	Model 1			Model 2		
	Coeff.	Std. err.	P-value	Coeff.	Std. err.	P-value
Constant	1.35	3.70	0.716	1.92	1.86	0.306
Low frequency	1.26	0.95	0.187	1.40	0.94	0.141
High frequency	3.49	1.04	0.001	3.54	1.03	0.001
Market leader	2.06	1.06	0.054	2.22	1.06	0.038
Agreeableness	−0.74	0.75	0.323	−0.37	0.71	0.600
Extraversion	0.94	0.75	0.216			
Openness to experience	0.31	0.77	0.689			
Neuroticism	−0.13	0.61	0.831			
Conscientiousness	−0.48	0.61	0.437			
Model	N = 109 Pseudo R ² = 0.030 p = 0.021			N = 109 Pseudo R ² = 0.025 p = 0.002		

Note: P-values refer to two-tailed coefficient tests.

TABLE A3 Classification of participants based on ROC responses.

Participant type	Frequency	Interpretation regarding Hypothesis 1
A	67 (32%)	Pro
B	24 (12%)	Contra
C	56 (27%)	Neutral
D	11 (5%)	Neutral
E	47 (23%)	Neutral
	205 (100%)	